THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of 101^3 is: 1

(a) 1030300 **(b)** 1030301 **(c)** 1030303 **(d)** 1030302

2. If the distance between $(x, 2)$ and $(1, 6)$ is 5 and $x > 1$, then $x =$ 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 5

3. If $x = 4$ is the zero of $kx - 28$, then $k =$ 1

(a) 7 **(b)** 8 **(c)** 6 **(d)** 9

4. $1.\overline{27}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{7}{6}$ **(b)** $\frac{13}{11}$ **(c)** $\frac{14}{11}$ **(d)** $\frac{15}{11}$

5. If $x + y = 10$ and $x^2 + y^2 = 58$, then $xy =$ 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 23

6. The fourth vertex of the parallelogram with consecutive vertices $(1, 1)$, $(4, 2)$, $(6, 5)$ (in order) is: 1

(a) $(-3, 4)$ **(b)** $(3, 4)$ **(c)** $(3, -4)$ **(d)** $(4, 3)$

7. If a linear polynomial $ax + b$ has zero 0, then: 1

(a) $a = b$ **(b)** $b = 0$ **(c)** $b = 1$ **(d)** $a = 0$

8. The value of $0.\overline{9}$ is: 1

(a) 0.9 **(b)** 1
(c) 0.99 **(d)** just less than 1

9. If $x - y = 3$ and $x^2 + y^2 = 65$, then $xy =$ 1

(a) 27 **(b)** 28 **(c)** 30 **(d)** 29

10. A point $P(x, 0)$ on the x-axis is 10 units from $(-3, 8)$ with $x > -3$. The value of x is: 1

(a) 4 **(b)** 5 **(c)** 2 **(d)** 3

11. For the pattern with nth term $6n - 1$, the sum of the 1st and 10th terms is: 1

(a) 66 **(b)** 63 **(c)** 65 **(d)** 64

12. The length of the repeating block of $\frac{1}{41}$ is: 1

(a) 5 **(b)** 4 **(c)** 7 **(d)** 6

13. **Assertion (A):** $103^2 = 10609$. 1
Reason (R): $(a+b)^2 = a^2 + 2ab + b^2$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The point $(0, -6)$ lies on the y-axis. 1
Reason (R): Every point on the x-axis has ordinate 0.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The graph of $y = 2x - 6$ cuts the x-axis at $(3, 0)$. 1
Reason (R): The graph of $y = 2x - 6$ cuts the y-axis at $(0, -6)$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $p(x) = 3x + 4$, find $p(p(x))$ and its value at $x = 2$. 2

17. Find the smallest natural number n such that $\frac{n}{420}$ has a terminating decimal expansion. 2

18. Find the value of 103^3 using an identity. 2

19. Find the distance between the points $(-7, 4)$ and $(2, -8)$. 2

20. Find the area of the triangle formed with the axes by the line $y = 2x + 14$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Factorise $x^2 - 5x + 4$ and use it to find the value when $x = 6$. 3

22. Find the length of the median AD of the triangle with $A(0, 0)$, $B(42, 0)$, $C(0, 56)$, where D is the midpoint of BC. 3

23. The nth term of a pattern is $14n + 3$. Find the number of terms that are less than 143 and the sum of the first two terms. 3

24. Given that $\sqrt{2}$ is irrational, prove that $3 + \sqrt{2}$ is irrational. 3

OR

Factorise $8x^3 - 27$. 3

25. Find the points on the x-axis whose distance from $A(-1, 24)$ is 26 units. 3

OR

Find the point of intersection of $y = 7x - 3$ and $y = 3x + 9$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Case study. A taxi company in Jaipur charges a fixed amount of Rs 50 plus Rs 12 for every kilometre travelled. The fare for x km is given by a polynomial $C(x)$. 5

(i) Write $C(x)$.

(ii) Find the fare for a 10 km ride.

(iii) For what distance is the fare Rs 230?

(iv) State the slope and the y-intercept of the graph of $y = C(x)$ and say what each means.

27. Simplify $(\sqrt{7} + \sqrt{5})^2 + (\sqrt{7} - \sqrt{5})^2$ and $(\sqrt{7} + \sqrt{5})(\sqrt{7} - \sqrt{5})$. Which are rational? 5

OR

If $x + y = 10$ and $x^2 + y^2 = 58$, find xy, $(x - y)^2$ and $x^2 - y^2$ given that $x > y$. 5

END OF QUESTION PAPER Total: 50 Marks

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